ED 419 Resampling Techniques and Statistical Computation

Markov Chain Monte Carlo in Epidemic Modeling

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BY

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Markov Chain Monte Carlo in Epidemic Modeling

A Real-Data Case Study Using India's COVID-19 Pre-Lockdown Data with R

Dataset & Methodology

Data: India MoHFW / Johns Hopkins University CSSE
Period: March 14–31, 2020 (pre-lockdown, 18 days)
Method: Adaptive Metropolis-Hastings MCMC (40,000 iters)
Code: R (`deSolve`, `coda`, `ggplot2`, `patchwork`)

Parameter	Post. Mean	2.5%	97.5%
β (transmission rate)	0.4675	0.433	0.508
γ (recovery rate)	0.1740	0.135	0.219
$R_0 = \beta/\gamma$	**2.72**	**2.31**	**3.21**
$1/\gamma$ (infectious period)	5.75 days	4.56	7.43

Acceptance rate: 18.1% Burn-in: 10,000 ESS(β) $\approx$ ESS(γ) $\approx$ 85

Abstract

Markov Chain Monte Carlo (MCMC) methods provide a principled framework for Bayesian inference over complex, analytically intractable posteriors. This report presents a complete real-world MCMC analysis applied to epidemic parameter estimation using actual India COVID-19 case data from the pre-lockdown phase (March 14–31, 2020). We fit a Susceptible-Infected-Recovered (SIR) model to 18 daily case observations from the Ministry of Health and Family Welfare (MoHFW), using an Adaptive Metropolis-Hastings sampler implemented in R. Our analysis recovers $R_0 = 2.72$ (95% CI: 2.31–3.21), consistent with published estimates for the original SARS-CoV-2 strain. We further reveal the *identifiability ridge* in the joint posterior of (β, γ), showing that $R_0 = \beta/\gamma$ is well-constrained by case data while the individual parameters remain partially unidentifiable. The full R code is provided and explained line by line.

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1. Introduction

During the early COVID-19 pandemic, a central challenge was estimating the basic reproduction number R_0 from sparse, noisy surveillance data. Classical approaches such as Maximum Likelihood Estimation (MLE) yield point estimates but cannot capture the full uncertainty structure of these parameters, which matters critically when hospital capacity and lockdown decisions hinge on worst-case scenarios rather than expected values.

Bayesian inference via MCMC offers a solution: instead of a single estimate, MCMC returns a *full posterior distribution*, capturing both epistemic uncertainty from limited data and structural uncertainty from partial parameter identifiability. This report presents a complete, reproducible R analysis on real data.

Why March 14–31, 2020? India's nationwide lockdown was announced on March 24. The 18 days in this window reflect unmitigated community transmission under natural contact patterns, making them ideal for identifying the intrinsic parameters β and γ of the SIR model.

The remainder of this report is structured as follows: Section 2 introduces the SIR model; Section 3 develops the Bayesian formulation; Section 4 derives the Metropolis-Hastings algorithm; Section 5 gives the full R implementation; Section 6 presents the results; and Section 8 concludes.

2. The SIR Compartmental Model

2.1. Model Structure

The SIR model partitions a closed population of size N into three compartments:

- $S(t)$: **Susceptible** — can contract the disease
- $I(t)$: **Infected** — currently infectious
- $R(t)$: **Recovered/Removed** — immune or deceased

with conservation $S(t) + I(t) + R(t) = N$ for all $t \geq 0$.

2.2. Governing Differential Equations

$$\frac{dS}{dt} = -\frac{\beta\, S\, I}{N} \tag{1}$$

$$\frac{dI}{dt} = \frac{\beta\, S\, I}{N} - \gamma\, I \tag{2}$$

$$\frac{dR}{dt} = \gamma\, I \tag{3}$$

where $\beta > 0$ is the **transmission rate** and $\gamma > 0$ the **recovery rate**, with $1/\gamma$ being the mean infectious period in days.

2.3. Basic Reproduction Number

$$R_0 = \frac{\beta}{\gamma} \tag{4}$$

Threshold Theorem

If $R_0 > 1$ the epidemic grows exponentially; if $R_0 < 1$ it dies out. The herd immunity threshold is $1 - 1/R_0$.

2.4. Observable: Daily New Infections

Surveillance data reports *daily new cases*, not compartment sizes. New infections equal the flux out of the susceptible compartment:

$$\text{new_cases}(t) = -\Delta S(t) = S(t) - S(t+1) \tag{5}$$

In R, this is computed numerically as `-diff(sol[, "S"])` from the `deSolve` ODE solution.

3. Bayesian Formulation

3.1. The Inference Problem

Let $\boldsymbol{\theta} = (\beta, \gamma)$ be the unknown parameters and $\mathbf{y} = (y_1, \ldots, y_T)$ the observed daily case counts. Bayes' theorem gives:

$$\underbrace{p(\boldsymbol{\theta} \mid \mathbf{y})}_{\text{posterior}} \propto \underbrace{p(\mathbf{y} \mid \boldsymbol{\theta})}_{\text{likelihood}} \cdot \underbrace{p(\boldsymbol{\theta})}_{\text{prior}} \tag{6}$$

The posterior is **analytically intractable** because the normalising constant requires integrating over the ODE-defined likelihood surface. MCMC circumvents this by sampling from $p(\boldsymbol{\theta} \mid \mathbf{y})$ without evaluating $p(\mathbf{y})$.

3.2. Likelihood

We model observation noise as Gaussian:

$$y_t \mid \boldsymbol{\theta} \sim \mathcal{N}\big(\hat{y}_t(\boldsymbol{\theta}),\ \sigma^2\big), \quad t = 1, \ldots, T \tag{7}$$

where $\hat{y}_t(\boldsymbol{\theta})$ is the ODE-predicted new case count and $\sigma = 8$ reflects the scale of reporting noise for the small early-phase counts. The log-likelihood is:

$$\log p(\mathbf{y} \mid \boldsymbol{\theta}) = -\frac{1}{2\sigma^2} \sum_{t=1}^{T} \big(y_t - \hat{y}_t(\boldsymbol{\theta})\big)^2 + \text{const} \tag{8}$$

3.3. Prior Distributions

Informative priors are specified from COVID-19 epidemiological knowledge:

$$\beta \sim \mathcal{N}(0.35,\ 0.08^2) \tag{9}$$
$$\gamma \sim \mathcal{N}(0.10,\ 0.025^2) \tag{10}$$

with the constraint $R_0 \in [1.0, 8.0]$ enforced as part of the prior support. The prior mean $\gamma = 0.10$ implies a 10-day mean infectious period, consistent with early COVID-19 estimates.

3.4. Log-Posterior

The unnormalised log-posterior — the only quantity required by MCMC:

$$\log p(\boldsymbol{\theta} \mid \mathbf{y}) \ = \ \log p(\mathbf{y} \mid \boldsymbol{\theta}) + \log p(\boldsymbol{\theta}) + \text{const} \tag{11}$$

4. Metropolis-Hastings Algorithm

4.1. Algorithm

The Metropolis-Hastings (MH) algorithm generates a Markov chain $\{\boldsymbol{\theta}^{(0)}, \boldsymbol{\theta}^{(1)}, \ldots\}$ whose stationary distribution is $p(\boldsymbol{\theta} \mid \mathbf{y})$. At each iteration i:

Step 1. Propose. Draw $\boldsymbol{\theta}^*$ from a Gaussian random walk:

$$\beta^* = \beta^{(i)} + \varepsilon_\beta, \quad \gamma^* = \gamma^{(i)} + \varepsilon_\gamma, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2_{\text{prop}})$$

Step 2. Acceptance ratio. For symmetric proposals the Hastings correction cancels:

$$\alpha = \min\Big(1,\ \exp\big[\log p(\boldsymbol{\theta}^* \mid \mathbf{y}) - \log p(\boldsymbol{\theta}^{(i)} \mid \mathbf{y})\big]\Big)$$

Step 3. Accept/Reject.

$$\boldsymbol{\theta}^{(i+1)} = \begin{cases} \boldsymbol{\theta}^* & \text{with prob. } \alpha \\ \boldsymbol{\theta}^{(i)} & \text{with prob. } 1-\alpha \end{cases}$$

Step 4. Iterate for $M = 40{,}000$ iterations, discard the first $B = 10{,}000$ as burn-in.

4.2. Proposal Adaptation

During burn-in the proposal scale is adapted via Robbins-Monro to target the 23.4% acceptance rate optimal for $d = 2$ dimensions:

$$\sigma^{(i+1)}_{\text{prop}} = \sigma^{(i)}_{\text{prop}} \cdot \exp\Big[0.2\,\big(\hat{\alpha}_i - 0.234\big)\Big] \tag{12}$$

where $\hat{\alpha}_i$ is the empirical acceptance rate at iteration i.

4.3. MAP Initialisation

The chain is initialised at the Maximum A Posteriori (MAP) estimate from a 25×25 grid search over $\beta \in [0.15, 0.65]$, $\gamma \in [0.04, 0.25]$. MAP: $(\hat{\beta}, \hat{\gamma}) = (0.4417, 0.1450)$, giving $\hat{R}_0 = 3.05$.

5. R Implementation

5.1. Dependencies and Data

```r
library(deSolve)     # ODE solver (lsoda)
library(coda)        # MCMC diagnostics (effectiveSize)
library(ggplot2)     # Visualisation
library(patchwork)   # Combine ggplot panels

# Real India COVID-19 daily new cases: March 14-31, 2020
# Source: India MoHFW / Johns Hopkins University CSSE
obs <- c(3, 5, 10, 13, 18, 24, 32, 42, 56, 73, 83, 87,
         90, 109, 117, 148, 176, 197)

T_len <- length(obs)    # 18 days of observation
N  <- 1380000000        # India population
I0 <- 3                 # Seed: 3 confirmed cases on March 14
```

Listing 1: Library imports and real COVID-19 data

5.2. SIR ODE and New-Cases Computation

```r
# SIR ordinary differential equation system
sir_ode <- function(time, state, parameters) {
  with(as.list(c(state, parameters)), {
    dS <- -beta * S * I / N
    dI <-  beta * S * I / N - gamma * I
    dR <-  gamma * I
    list(c(dS, dI, dR))
  })
}

# Simulate SIR and return daily new infections = -Delta S
simulate <- function(beta, gamma) {
  init   <- c(S = N - I0, I = I0, R = 0.0)
  times  <- seq(0, T_len, by = 1)
  params <- c(beta = beta, gamma = gamma)

  out <- tryCatch(
    ode(y = init, times = times, func = sir_ode,
        parms = params, method = "lsoda"),
    error = function(e) NULL
  )

  if (is.null(out)) return(NULL)
  pmax(-diff(out[, "S"]), 0)    # new infections on each day
}
```

Listing 2: SIR ODE and simulation function

5.3. Log-Likelihood, Prior, and Log-Posterior

```
sigma <- 8.0   # Observation noise (scale of reporting uncertainty)

# Gaussian log-likelihood: Eq. (5)
log_lik <- function(pred, obs) {
  if (is.null(pred) || any(pred <= 0)) return(-Inf)
  -0.5 * sum(((obs - pred) / sigma)^2)
}

# Informative prior: Eq. (6)-(7)
# beta  ~ N(0.35, 0.08^2) from contact-rate epidemiology
# gamma ~ N(0.10, 0.025^2) from ~10-day mean infectious period
log_prior <- function(beta, gamma) {
  if (beta <= 0 || gamma <= 0) return(-Inf)
  R0 <- beta / gamma
  if (R0 < 1.0 || R0 > 8.0) return(-Inf)   # soft R0 constraint
  lp_b <- -0.5 * ((beta  - 0.35) / 0.08)^2
  lp_g <- -0.5 * ((gamma - 0.10) / 0.025)^2
  lp_b + lp_g
}

# Unnormalised log-posterior: Eq. (8)
log_post <- function(beta, gamma) {
  lp <- log_prior(beta, gamma)
  if (!is.finite(lp)) return(-Inf)
  pred <- simulate(beta, gamma)
  lp + log_lik(pred, obs)
}
```

Listing 3: Statistical model: likelihood, prior, and posterior

5.4. MAP via Grid Search

```
cat("Step 1: Grid search for MAP...\n")
best_lp <- -Inf; best_b <- 0.35; best_g <- 0.10

beta_grid  <- seq(0.15, 0.65, length.out = 25)
gamma_grid <- seq(0.04, 0.25, length.out = 25)

for (b in beta_grid) {
  for (g in gamma_grid) {
    lp <- log_post(b, g)
    if (lp > best_lp) {
      best_lp <- lp; best_b <- b; best_g <- g
    }
  }
}
cat(sprintf("MAP: beta=%.4f, gamma=%.4f, R0=%.2f\n",
```

```
            best_b, best_g, best_b / best_g))
```

Listing 4: Grid search for MAP initialisation

5.5. Adaptive Metropolis-Hastings Sampler

```
set.seed(7)
n_iter  <- 40000
burn_in <- 10000
prop_sd <- c(0.010, 0.004)   # initial proposal SD for (beta, gamma)

# Initialise at MAP
beta_c  <- best_b; gamma_c <- best_g; lp_c <- best_lp
chain_b <- numeric(n_iter);  chain_g <- numeric(n_iter)
accepted <- 0

for (i in 1:n_iter) {
  # Step 1: Gaussian random-walk proposal
  bp <- beta_c  + rnorm(1, 0, prop_sd[1])
  gp <- gamma_c + rnorm(1, 0, prop_sd[2])

  # Step 2: log acceptance ratio (symmetric q cancels)
  lp_p      <- log_post(bp, gp)
  log_alpha <- lp_p - lp_c

  # Step 3: accept / reject
  if (log(runif(1) + 1e-300) < log_alpha) {
    beta_c <- bp; gamma_c <- gp; lp_c <- lp_p
    accepted <- accepted + 1
  }
  chain_b[i] <- beta_c; chain_g[i] <- gamma_c

  # Robbins-Monro proposal adaptation (during burn-in)
  if (i > 500 && i < burn_in && i %% 500 == 0) {
    ar      <- accepted / i
    prop_sd <- prop_sd * exp(0.2 * (ar - 0.234))
    prop_sd <- pmax(pmin(prop_sd, 0.2), 1e-4)
  }

  # Progress report
  if (i %% 10000 == 0) {
    cat(sprintf("Iter %5d | beta=%.5f gamma=%.5f | Accept: %.1f%%\n",
                i, beta_c, gamma_c, accepted / i * 100))
  }
}

# Discard burn-in
post_b  <- chain_b[(burn_in + 1):n_iter]
post_g  <- chain_g[(burn_in + 1):n_iter]
post_R0 <- post_b / post_g

# Diagnostics
cat(sprintf("Acceptance rate: %.1f%%\n", accepted / n_iter * 100))
```

```r
ci_b  <- quantile(post_b,  c(0.025, 0.975))
ci_g  <- quantile(post_g,  c(0.025, 0.975))
ci_R0 <- quantile(post_R0, c(0.025, 0.975))
cat(sprintf("beta  = %.5f  [95%% CI: %.5f-%.5f]\n",
            mean(post_b),  ci_b[1],  ci_b[2]))
cat(sprintf("gamma = %.5f  [95%% CI: %.5f-%.5f]\n",
            mean(post_g),  ci_g[1],  ci_g[2]))
cat(sprintf("R0    = %.3f    [95%% CI: %.3f-%.3f]\n",
            mean(post_R0), ci_R0[1], ci_R0[2]))
cat(sprintf("ESS(beta)=%.0f  ESS(gamma)=%.0f\n",
            effectiveSize(post_b), effectiveSize(post_g)))
```

Listing 5: Adaptive Metropolis-Hastings MCMC (40,000 iterations)

5.6. Posterior Predictive and Visualisation

```r
# Posterior predictive: simulate 500 draws from the posterior
idx        <- sample(length(post_b), 500, replace = FALSE)
curves     <- lapply(idx, function(j) simulate(post_b[j], post_g[j]))
curves_mat <- do.call(rbind, Filter(Negate(is.null), curves))
mean_pred  <- colMeans(curves_mat)
lower_pred <- apply(curves_mat, 2, quantile, probs = 0.025)
upper_pred <- apply(curves_mat, 2, quantile, probs = 0.975)

# Shared dark ggplot2 theme
dark_theme <- theme_minimal() + theme(
  plot.background  = element_rect(fill = "#0D1B2A", color = NA),
  panel.background = element_rect(fill = "#132236", color = NA),
  panel.grid.major = element_line(color = "#1C3F6E", linewidth = 0.5),
  text             = element_text(color = "#8FA8BE"),
  axis.text        = element_text(color = "#8FA8BE"),
  plot.title       = element_text(color = "white", face = "bold", size = 13),
  legend.background = element_rect(fill = "#132236", color = NA),
  legend.text      = element_text(color = "white")
)

# Trace plots (beta and gamma stacked with patchwork)
p_trace_b <- ggplot(data.frame(iter = 1:n_iter, val = chain_b),
                    aes(x = iter, y = val)) +
  geom_line(color = "#00A8CC", alpha = 0.88, linewidth = 0.45) +
  geom_vline(xintercept = burn_in, color = "white",
             linetype = "dashed", linewidth = 0.8) +
  geom_hline(yintercept = mean(post_b), color = "yellow",
             linetype = "dotted", linewidth = 0.8) +
  labs(title = "MCMC Trace Plots -- India COVID-19 Pre-Lockdown Phase",
       y = "beta (Transmission Rate)", x = NULL) +
  dark_theme

p_trace_g <- ggplot(data.frame(iter = 1:n_iter, val = chain_g),
                    aes(x = iter, y = val)) +
  geom_line(color = "#FF8C42", alpha = 0.88, linewidth = 0.45) +
  geom_vline(xintercept = burn_in, color = "white",
             linetype = "dashed", linewidth = 0.8) +
```

```r
  geom_hline(yintercept = mean(post_g), color = "yellow",
             linetype = "dotted", linewidth = 0.8) +
  labs(y = "gamma (Recovery Rate)", x = "MCMC Iteration") +
  dark_theme

fig_trace <- p_trace_b / p_trace_g

# Posterior histogram helper
plot_post <- function(data, title, col, xlab) {
  mn <- mean(data); lo <- quantile(data, 0.025); hi <- quantile(data, 0.975)
  ggplot(data.frame(val = data), aes(x = val)) +
    geom_histogram(aes(y = after_stat(density)), bins = 55,
                   fill = col, alpha = 0.83) +
    geom_vline(xintercept = mn, color = "white", linewidth = 1.2) +
    geom_vline(xintercept = lo, color = "#aaaaaa",
               linewidth = 0.8, linetype = "dashed") +
    geom_vline(xintercept = hi, color = "#aaaaaa",
               linewidth = 0.8, linetype = "dashed") +
    labs(title = title, x = xlab, y = "Density") +
    dark_theme + theme(plot.title = element_text(color = col, size = 10.5))
}

fig_post <- (plot_post(post_b,  "beta", "#00A8CC", "beta") |
             plot_post(post_g,  "gamma", "#FF8C42", "gamma") |
             plot_post(post_R0, "R0 = beta/gamma", "#5CE1E6", "R0")) +
  plot_annotation(title = "Posterior Distributions -- MCMC Estimates",
                  theme = dark_theme)

# Observed vs posterior predictive
df_fit <- data.frame(day  = 0:(T_len-1), obs  = obs,
                     mean = mean_pred, lo = lower_pred, hi = upper_pred)

fig_fit <- ggplot(df_fit, aes(x = day)) +
  geom_ribbon(aes(ymin = lo, ymax = hi),
              fill = "#00A8CC", alpha = 0.28) +
  geom_line(aes(y = mean), color = "#00A8CC", linewidth = 1.2) +
  geom_point(aes(y = obs), color = "#FF8C42", size = 3) +
  geom_vline(xintercept = 11, color = "white",
             linetype = "dotted", linewidth = 0.8, alpha = 0.6) +
  annotate("text", x = 11.4, y = max(obs) * 0.85,
           label = "Lockdown\nMar 24", color = "#aaaaaa", size = 3) +
  labs(title = "India COVID-19: Observed vs MCMC Posterior Predictive",
       subtitle = "(March 14-31, 2020 | 18 days | Real MoHFW data)",
       x = "Day", y = "Daily New Cases") + dark_theme

# Joint posterior -- identifiability ridge
df_joint <- data.frame(
  beta  = post_b[ seq(1, length(post_b), by = 4)],
  gamma = post_g[ seq(1, length(post_g), by = 4)],
  R0    = post_R0[seq(1, length(post_R0), by = 4)]
)
fig_joint <- ggplot(df_joint, aes(x = beta, y = gamma, color = R0)) +
  geom_point(alpha = 0.40, size = 1) +
  scale_color_viridis_c(option = "plasma", name = "R0") +
```

```
  annotate("point", x = mean(post_b), y = mean(post_g),
           color = "white", shape = 8, size = 5) +
  labs(title = "Joint Posterior: beta vs gamma (colored by R0)",
       subtitle = "Diagonal ridge = partial identifiability",
       x = "beta (transmission rate)", y = "gamma (recovery rate)") +
  dark_theme
```

Listing 6: Posterior predictive sampling and ggplot2 plots

6. Results and Interpretation

6.1. MCMC Convergence

Table 1: MCMC convergence diagnostics

Diagnostic	Value	Assessment
Total iterations	40,000	—
Burn-in discarded	10,000	—
Post-burn-in samples	30,000	—
Acceptance rate	18.1%	Good (target 15–25%)
ESS(β)	≈ 85	Adequate
ESS(γ)	≈ 85	Adequate
Trace stationarity	Yes	Visual inspection

6.2. Posterior Parameter Estimates

Table 2: Posterior summary statistics

Parameter	Mean	Median	SD	2.5%	97.5%
β	0.4675	0.4682	0.020	0.433	0.508
γ	0.1740	0.1748	0.021	0.135	0.219
R_0	2.720	2.689	0.233	2.312	3.211
$1/\gamma$ (days)	5.75	5.72	0.69	4.56	7.43

Key Finding

$R_0 = 2.72$ (95% CI: 2.31–3.21) is **consistent with published meta-analytic estimates** for original SARS-CoV-2 in India (range: 2.0–3.5), validating the Bayesian MCMC approach on 18 days of real case data.

6.3. Identifiability Ridge

The joint posterior of (β, γ) shows a pronounced *diagonal ridge*: infinitely many (β, γ) pairs share the same exponential growth rate $r = \beta - \gamma$, creating the constraint:

$$\gamma = \beta - r \quad \Longrightarrow \quad \text{diagonal ridge in the } (\beta, \gamma) \text{ plane.}$$

Implication. $R_0 = \beta/\gamma$ is more tightly constrained (CI width 0.90) than β or γ individually (CI widths 0.075 and 0.084). Policy should be based on the R_0 posterior, not on individual parameters.

6.4. Comparison with Alternative Methods

Table 3: Inference methods compared

Method	Full Posterior	Prior	Uncertainty	Scalability
Least Squares	No	No	None	High
MLE	No	No	Fisher Info CI	High
MCMC (this work)	Yes	Yes	Full posterior	Moderate
HMC / Stan	Yes	Yes	Full posterior	Higher

7. Discussion

Strengths. Full uncertainty quantification allows computing tail probabilities directly (e.g., $\Pr(R_0 > 3)$ from the posterior). Informative priors regularise inference from small datasets. Posterior predictive checks validate model adequacy.

Limitations. The SIR model assumes homogeneous mixing and constant β, neither of which holds in reality. India's testing capacity was very limited in March 2020, implying significant under-reporting. The ESS of ≈ 85 is adequate for point estimates but modest; for tail probabilities, Hamiltonian Monte Carlo (e.g., via `rstan`) would be preferable.

8. Conclusion

We presented a complete Bayesian MCMC pipeline for SIR epidemic parameter estimation, implemented in R and applied to real India COVID-19 data. The Adaptive Metropolis-Hastings sampler recovered $R_0 = 2.72$ (95% CI: 2.31–3.21), consistent with the published literature. The identifiability ridge in the joint posterior revealed that epidemic case-count data constrains R_0 far more tightly than β or γ individually — an insight that point-estimate methods cannot provide.

Final Summary

Data:	18 days, India MoHFW / JHU CSSE (March 14–31, 2020)
Method:	Adaptive Metropolis-Hastings, 40,000 iterations
$\hat{R}_0$**:**	2.72 [95% CI: 2.31–3.21]
Acc. rate:	18.1% (target: 15–25%)
ESS:	≈ 85 per parameter

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